Day 11: Inference for MLR

`bikes` data

Goal: Predict _____ based on `temp_actual` and `weekend`

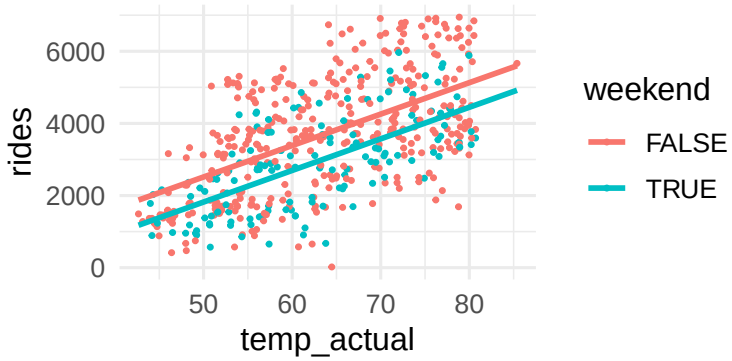
term	estimate	std. error	t	p-value
Intercept	-1843.867	415.700	-4.44	<0.001
temp_actual	87.173	6.439	13.54	<0.001
weekendTRUE	-714.938	795.698	-0.90	0.369
temp_actual:weekendTRUE	0.411	12.539	0.03	0.974

Warm up:

Write out the theoretical model from the table above

Write out the fitted model for weekends and weekdays separately

Do we need a different-slopes model?



Answer based on the graph:

Answer based on the table:

T-test for a single coefficient

Hypotheses: $H_0 : \beta_j = \#$ vs. $H_a : \beta_j \neq, <, \text{or} > \#$

Test statistic: $t = \dfrac{\hat{\beta}_j - \#}{SE(\hat{\beta}_j)}$

Reference distribution: t distribution with $df = n - (p + 1)$, where p = number of predictors in the model (an interaction counts as one)

p-value: area in the tail(s) specified by H_a

CI for a single coefficient:

$\hat{\beta}_j \pm t^*_{n-(p+1)} \cdot SE(\hat{\beta}_j)$

Example: Use the coefficient table at the top of the notes and `qt(0.975, df = 496)` to calculate a 95% confidence interval for the slope of `temp_actual` (the slope for weekdays).

Do we need weekend at all?

Answer:

Now drop only the interaction between weekend and `temp_actual` (the parallel-lines model):

term	estimate	std. error	t	p-value
Intercept	-1850.776	357.953	-5.17	<0.001
temp_actual	87.281	5.520	15.81	<0.001
weekendTRUE	-689.179	124.639	-5.53	<0.001

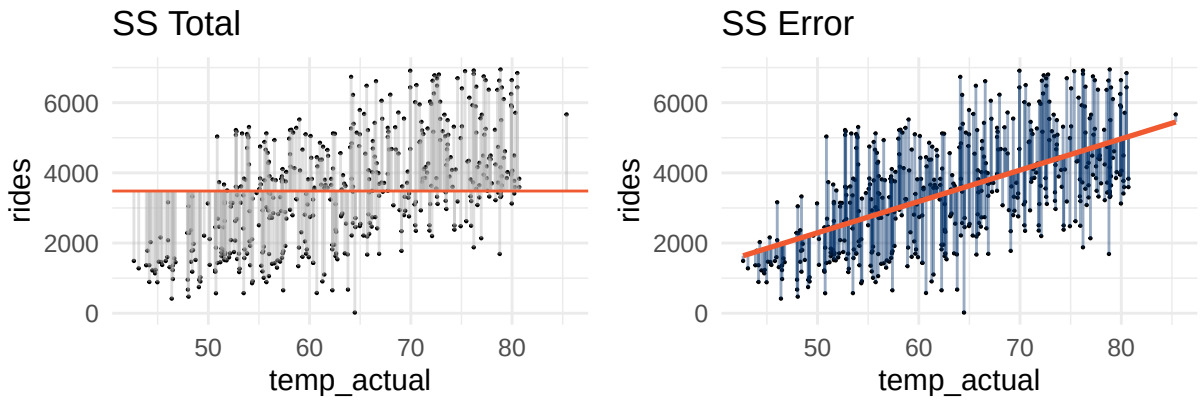
What changed about weekend?

> **Caution!**
>
> $\hat{\beta}$'s and p-values are highly _____ on each other! _____ β's might become _____ after removing or adding another term

A t-test can only ask about **one** coefficient at a time, given everything else in the model. We need a test that asks about several coefficients at once

Explained variability

Idea: does the **full** model reduce the error (SSE) by *more than chance alone would*?



Comparing nested models: the Extra Sum of Squares F Test

Full Model: $\mu\{Y|X\} = \beta_0 + \beta_1 \text{temp} + \beta_2 \text{weekend} + \beta_3 \text{temp} \times \text{weekend}$

Reduced Model: $\mu\{Y|X\} = \beta_0 + \beta_1 \text{temp}$

The reduced model is the full model with some β's set to 0, so we say the models are **nested**.

- H_0: _____
- H_a : _____

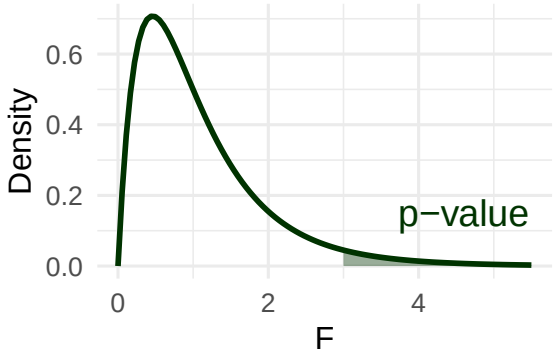
$$F = \frac{(\text{SSE}_{\text{reduced}} - \text{SSE}_{\text{full}})/d}{\text{MSE}_{\text{full}}} = \frac{\text{how much error drops per } \beta}{\text{estimate of } \sigma^2}$$

- If H_0 is true, both parts estimate σ^2, so $F \approx$ _____
- If the extra terms really help, F is much _____ than 1

where

- $d = \text{df}_{\text{reduced}} - \text{df}_{\text{full}}$ is the number of β's being tested
- $\text{df}_i = n - (p_i + 1)$ is the error d.f. for model i
- $\text{MSE}_{\text{full}} = \text{SSE}_{\text{full}}/\text{df}_{\text{full}}$.

The F statistic follows an F distribution with d and df_{full} degrees of freedom. It is right-skewed, and only large values of F are evidence against H_0, so the p-value is the **upper tail**:



Upper-tail area in R: `pf(stat, df1, df2, lower.tail = FALSE)`

Model	SSE	df
Reduced: temp	826,326,178	498
Full: temp * weekend	778,436,469	496

Example: Calculate the F statistic.

```
pf(f_stat, df1 = 2, df2 = 496, lower.tail = FALSE)
## [1] 3.703116e-07
```

→ We _____ H_0 and conclude:

Extra Sum of Squares F test in R

```
full <- lm(rides ~ temp_actual * weekend, data = bikes)
reduced <- lm(rides ~ temp_actual, data = bikes)
anova(reduced, full)
```

```
Analysis of Variance Table

Model 1: rides ~ temp_actual
Model 2: rides ~ temp_actual * weekend
  Res.Df       RSS Df Sum of Sq      F    Pr(>F)
1    498 826326178
2    496 778436469  2  47889709 15.257 3.713e-07 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
Analysis of Variance Table

Model 1: rides ~ temp_actual
Model 2: rides ~ temp_actual * weekend
  Res.Df       RSS Df    Sum of Sq      F               Pr(>F)
1    df1      SSE1
2    df2      SSE2  d    SSE1-SSE2  (SumSq/d)/(SSE2/df2) p-val
```

Resolving the puzzle

Earlier, the t-tests in the full model said neither weekend term was useful, but weekend was highly significant once the interaction was dropped. The F-test lets us ask each question directly:

Question	Reduced	Full	d	F	p-value
Different slopes needed?	temp + weekend	temp * weekend	1	0.00107	0.974
Weekend shift needed (no interaction)?	temp	temp + weekend	1	30.6	<0.001
Weekend needed at all?	temp	temp * weekend	2	15.3	<0.001

- One tool answers all three questions
- When $d = 1$, the F-test and the t-test are the same test: $F = $ _____
- The t-tests couldn't answer the *joint* question (row __)

`anova()` on a single model

```
anova(bikes_mlr)
```

```
Analysis of Variance Table

Response: rides
                     Df    Sum Sq   Mean Sq  F value    Pr(>F)
temp_actual           1 411018312 411018312 261.8905 < 2.2e-16 ***
weekend               1  47888023  47888023  30.5130 5.361e-08 ***
temp_actual:weekend   1      1686      1686   0.0011    0.9739
Residuals           496 778436469   1569428
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

- Each row adds one term to the model, in order, and tests whether it improves the fit _____ the terms above it
- The **last row** is the extra sums of squares F-test for the last term (here, the interaction) and matches `anova(bikes_mlr2, bikes_mlr)`

Reading R output

```
lm(formula = rides ~ temp_actual * weekend, data = bikes)

Coefficients:
                         Estimate Std. Error t value Pr(>|t|)
(Intercept)             -1843.867    415.700  -4.436 1.13e-05 ***
temp_actual                87.173      6.439  13.538  < 2e-16 ***
weekendTRUE              -714.938    795.698  -0.899    0.369
temp_actual:weekendTRUE     0.411     12.539   0.033    0.974

Residual standard error: 1253 on 496 degrees of freedom
Multiple R-squared:  0.3709,    Adjusted R-squared:  0.3671
F-statistic: 97.47 on 3 and 496 DF,  p-value: < 2.2e-16
```

The **F-statistic** at the bottom is the same kind of test: the full model vs. an *intercept-only* model.

H_0: _____ (none of the predictors help)

Adjusted R^2

$$R^2 = 1 - \frac{SSE}{SST} \qquad\qquad R^2_{adj} = 1 - \frac{SSE/(n-p-1)}{SST/(n-1)} = 1 - \frac{MSE}{MST}$$

where p is the number of predictors in the model.

Example: Look at the adjusted R^2 column. Which model has the largest adjusted R^2? Did adding the interaction help?

Model	R^2	Adjusted R^2
temp	0.3322	0.3308
temp + weekend	0.3709	0.3683
temp * weekend	0.3709	0.3671
temp + 5 random noise variables	0.3392	0.3312

- R^2 **never** goes _____ when predictors are added, even for pure _____
- R^2_{adj} **could** _____ if unnecessary predictors are added
- Better to use R^2_{adj} than R^2 to *compare* models, but neither is a formal test

Takeaways

- A t-test asks about **one** coefficient, *given* everything else in the model
- To test **several** coefficients at once, compare nested models with an **extra sums of squares F-test**
- $F = \dfrac{(\text{SSE}_{\text{reduced}} - \text{SSE}_{\text{full}})/d}{\text{MSE}_{\text{full}}}$: error reduction per extra β, relative to the noise
- R^2 always goes up; the F-test tells us whether the improvement is *real*